

Effects of EPX, IMD and BOS on *A. caliginosa* - Preliminary tests

Protocol

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Objective : To identify the concentration ranges for each molecule to test that will enable the construction of dose-response curves for growth and reproduction.

1 Generalities

1.1 Earthworms

Earthworms for these preliminary tests are *A. icterica* individuals fetch in a meadow in Versailles (Les Closeaux). *A. caliginosa* were unavailable and *A. icterica*, a similar species, is much easier to fetch in Versailles. The use of this species will still allow for the identification of concentration ranges to test for the real EC₅₀ tests.

1.2 Design

Table 1.1: Primary solutions concentrations for each molecule in acetone

molecule	concentration_p1	concentration_p2
EPX	3000	600
IMD	300	60
BOS	3000	600

Cosms are made with 200 g of Closeaux soil and earthworms are fed with 3 g/ind/14d.

Each modality is made with 1 replicates for growth and 2 replicates for reproduction.

The growth test lasts 28 days and the reproduction test lasts 28 days.

Table 1.2: Tested concentrations for each molecule (mg/kg)

EPX	IMD	BOS
1e-02	1e-03	1e-02
1e-01	1e-02	1e-01
1e+00	1e-01	1e+00
1e+01	1e+00	1e+01
1e+02	1e+01	1e+02

15 concentrations tested.

For the growth test, one replicate correspond to 2 individuals per cosm. For the reproduction test, one replicate correspond to 2 individuals per cosm.

1.3 Number of individuals and number of cosms

Table 1.3: Number of needed individuals

Experience	EPX	IMD	BOS	Total
Growth	10	10	10	30
Reproduction	20	20	20	60
Total	30	30	30	90

Table 1.4: Number of needed cosms

Experience	Nb_cosm
Growth	15
Reproduction	30
Total	45

1.4 Soil and horse dung needs

Cosms preparation needs 9 kg of Closeaux soil. 540 g of horse dung are needed in total.

2 Protocol

2.1 Calculations

2.1.1 Preparations for EPX

Table 2.1: Soil preparation for each EPX concentration

Concentration (mg/kg)	Q_{EPX} 1conc (mg)	Q_{tot} sol (g)	Veau (CRE 70%)	Vsol.p1 (μ L)	Vsol.p2 (μ L)	Choice Sol.p
0.01	0.006	600	823.2	2	10	2
0.10	0.060	600	823.2	20	100	2
1.00	0.600	600	823.2	200	1000	2
10.00	6.000	600	823.2	2000	10000	2
100.00	60.000	600	823.2	20000	100000	1

20 mL of primary solution n°1 are needed.

11.11 mL of primary solution n°2 are needed..

25 mL of primary solution n°1 are prepared with 75 mg of EPX in Acetone. 20 mL of primary solution n°2 are prepared with 4 mL of the primary solution n°1 in Acetone.

2.1.2 Preparations for IMD

Table 2.2: Soil preparation for each IMD concentration

Concentration (mg/kg)	Q_{IMD} 1conc (mg)	Q_{tot} sol (g)	Veau (CRE 70%)	Vsol.p1 (μ L)	Vsol.p2 (μ L)	Choice Sol.p
0.001	0.0006	600	823.2	2	10	2
0.010	0.0060	600	823.2	20	100	2
0.100	0.0600	600	823.2	200	1000	2
1.000	0.6000	600	823.2	2000	10000	2
10.000	6.0000	600	823.2	20000	100000	1

20 mL of primary solution n°1 are needed.

11.11 mL of primary solution n°2 are needed.

100 mL of the primary solution n°1 are prepared with 30 mg of IMD (kept in the fridge à 4°C).

20 mL of primary solution n°2 are prepared with 4 mL of the primary solution n°1 in Acetone.

2.1.3 Preparations for BOS

Table 2.3: Soil preparation for each BOS concentration

Concentration (mg/kg)	Q_{BOS} lconc (mg)	Q_{tot} sol (g)	Veau (CRE 70%)	Vsol.p1 (μ L)	Vsol.p2 (μ L)	Choice Sol.p
0.01	0.006	600	823.2	2	10	2
0.10	0.060	600	823.2	20	100	2
1.00	0.600	600	823.2	200	1000	2
10.00	6.000	600	823.2	2000	10000	2
100.00	60.000	600	823.2	20000	100000	1

20 mL of primary solution n°1 are needed.

11.11 mL of primary solution n°2 are needed..

25 mL of primary solution n°1 are prepared with 75 mg of BOS in Acetone. 20 mL of primary solution n°2 are prepared with 4 mL of the primary solution n°1 in Acetone.

2.2 D-4 - Soil preparation

In the morning, around 6 kg of soil are put in steel recipient in a thermoregulated chamber at 110°C.

In the afternoon, in 15 inox recipient, 600 g of this soil are weighted and 823.2 mL of ISO water are added (do not mix !).

2.3 D-3 - Soil contamination

Cosms left from the first try are taken out from the refrigerated chamber at -20°C and are put in the thermoregulated chamber at 18°C.

Primary solutions are prepared.

For each molecule and each concentrations :

- The volume given by Table 2.3, Table 2.2 or Table 2.1 is put in a 10 mL beaker
- Acetone is added to have a volume of around 10 mL

- This solution is directly put on the soil while mixing with a syringe
- The cosms for the given concentration are prepared with 255 g of this soil preparation
- Cosms are left open under a hood.

3 D0 - Eathworms introduction

Horse dung is added and mixed to each cosm : 3 g/ind (dry weight) with its WHC adjust to 70%.

Juveniles earthworms are individually weighted and put into their cosms by two. For each reproduction cosm, two adults are weighted together and then put into the cosm.

The weight of the cosms are recorded to be able to adjust the soil humidity over time. s